

A Fact-Checking Framework with Denoising Evidence Retrieval and LLM-Based Debate Verification

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Abstract

The rapid spread of misinformation on social media has underscored the importance of automatic fact-checking. Existing fact-checking pipelines typically rely on multi-stage frameworks involving evidence retrieval and claim verification. However, these methods face two major challenges: (1) the retrieval process often introduces noisy evidence, which compromises the reliability of the final veracity prediction; and (2) the verification models may overlook critical factual details, resulting in hallucinated conclusions. To address these issues, we propose a fact-checking framework SLED with Self-supervised denoising evidence retrieval and LLM-Enhanced Debate-based verification. In the retrieval stage, SLED leverage trained verifier to assess *credibility* and *necessity* of retrieved evidence, enabling the elimination of noisy evidence. In the verification stage, SLED prompts the LLM to generate dual-perspective reasoning and simulates a multi-agent debate, followed by distillation into a lightweight model for final veracity prediction. Experiments on CHEF and HOVER datasets demonstrate that SLED achieves the state-of-the-art results in complex fact verification scenarios.

CCS Concepts

• Computing methodologies → Natural language processing.

Keywords

Natural Language Inference, Fact-checking, Denoising, Agent

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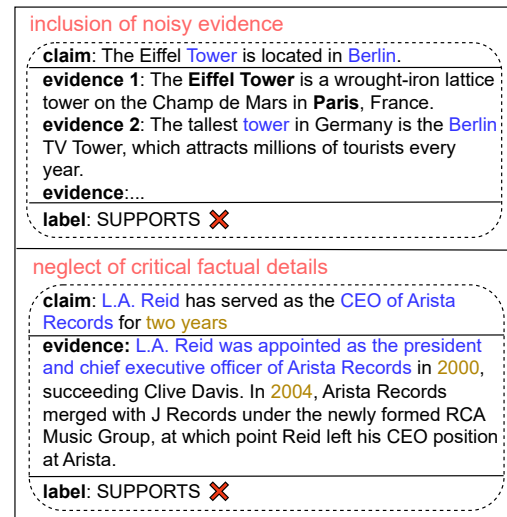


Figure 1: Fact-checking involves evidence retrieval and claim verification. Existing methods may yield inaccurate veracity prediction due to the inclusion of noisy evidence or the neglect of critical factual details.

1 Introduction

The rise of social media and large language models (LLMs) has led to a rapid spread of both accurate and misleading information online [1, 2, 11, 16, 33, 35]. This has raised interest in automatic fact-checking, which aims to determine the veracity of claims using external evidence [6, 19, 57].

Unlike earlier fact-checking methods that rely mainly on the semantic or linguistic patterns of claims [44, 45], Thorne et al. [38] introduces a structured pipeline consisting of evidence retrieval and claim verification, which has been proven effective, as each stage addresses a distinct subtask and contributes to the overall

reliability of fact-checking. However, two major limitations hinder current approaches.

On the one hand, in the evidence retrieval stage, existing approaches [30, 39, 47, 49] often tend to retrieve as much evidence as possible for multi-hop reasoning, which inevitably introduces noisy information and may undermine the accuracy of the final veracity prediction. As illustrated in Figure 1, the retrieved evidence related to the terms “tower” and “Berlin” significantly influence the final veracity prediction of the claim. Therefore, effectively retrieving relevant, accurate, and noise-free evidence with minimal human annotation is critical for enhancing fact-checking performance.

On the other hand, recent approaches have leveraged the growing reasoning capabilities of large language models (LLMs) to achieve promising results in claim verification [17, 18, 27, 52]. However, these methods are prone to hallucinations due to the omission of critical information in the claim, which undermines the accuracy of veracity prediction. For example, as depicted in Figure 1, given the claim, a model may predict SUPPORTS based on partial evidence indicating Reid’s affiliation with Arista, while ignoring the temporal scope implied by “two years”, thus arriving at an incorrect judgment. Therefore, an important challenge in claim verification is how to fully leverage the retrieved evidence to make more comprehensive and accurate veracity prediction.

To address the challenges posed by inclusion of noisy evidence in the retrieval process and the hallucination phenomenon arising from the neglect of factual details during verification, we propose SLED, a fact-checking framework with Self-supervised denoising evidence retrieval and LLM-Enhanced Debate-based verification. In the evidence retrieval stage, we first train a verifier using the ground-truth veracity labels of claims, and then leverage it to assess the *credibility* and *necessity* of each retrieved evidence, ensuring that each piece of evidence independently contributes to truthfulness verification while also maintaining its necessity, which ultimately yields an accurate, noise-free set of evidence. In the verification stage, the supporting and refuting agents independently generate reasoning steps from their respective stances. They then engage in a debate by posing questions and challenges targeting each other’s factual details. A judge agent summarizes the debate process, and a distilled small model leverages this summary to make the final veracity prediction.

We evaluate our approach on two challenging fact-checking datasets, CHEF [10] and HOVER [13]. Experimental results show that our method achieves state-of-the-art performance, demonstrating its effectiveness in complex claim verification scenarios. The main contributions of our works can be summarized as follows:

- We introduce a self-supervised denoising evidence retrieval model that scores candidate evidence based on *credibility* and *necessity*, enabling high-quality evidence selection.
- We design a debate-guided multi-agent reasoning framework in which two agents argue from opposing perspectives and a judge agent determines the verdict. To enhance scalability, we distill this process into a lightweight model that evaluates both sides’ reasoning to improve verification performance.
- Experimental results show that our method achieves state-of-the-art performance on CHEF and HOVER, demonstrating its effectiveness in complex claim verification scenarios.

2 Related Works

Early studies approached fact verification from a claim-only perspective, predicting veracity solely based on linguistic features. Traditional methods relied on bag-of-words [44], convolutional neural networks [15], graph-based propagation [45, 50], keyword cues [12, 29], or speaker stance modeling [32]. However, such approaches suffer from limited interpretability and poor generalization, and often fail against carefully crafted misinformation [48, 54].

Later works incorporate evidence retrieval to address more sophisticated misinformation. Hassan et al. [8], Thorne and Vlachos [37], Vlachos and Riedel [40] formalize fact-checking as a pipeline of claim identification, evidence retrieval, and verification. Evidence quality is improved through entity-based retrieval [7], DQN-based selection [42], autoregressive reranking [14], knowledge integration [58], and unified evidence enhancement [47]. MRR [56] further mitigates intent drift via generative multi-hop retrieval and hierarchical reasoning. Nevertheless, multi-hop retrieval remains vulnerable to noisy evidence, which can degrade verification accuracy.

For claim verification, prior work explores hierarchical attention [20], joint representation learning [31], debiasing or causal inference techniques [54], and dual-view frameworks [46]. Despite improved reasoning ability, these methods may overlook critical factual information, leading to incorrect veracity predictions.

With the rise of large language models (LLMs), recent studies leverage their internal knowledge and reasoning capabilities for fact verification. Approaches such as ProtoCo [53], HiSS [55], ProgramFC [27], Self-Checker [17], and ARG [9] improve verification through prompting, claim decomposition, modular reasoning, or rationale guidance. Other works further enhance performance via iterative retrieval [41, 56] or multi-agent collaboration [21]. However, LLM-based methods remain prone to hallucinations when critical evidence is overlooked.

Despite these advances, existing LLM-based approaches often suffer from hallucinations due to overlooking critical information, which undermines veracity prediction. To mitigate this, our method employs two collaborative agents that engage in debate from different perspectives, effectively reducing hallucination in LLM reasoning.

3 Method

As illustrated in Figure 2, the proposed SLED framework consists of two main components: (1) Evidence Retrieval: a self-supervised approach evaluates the credibility and necessity of retrieved open-domain evidence to filter relevant content; and (2) Claim Verification: a multi-agent debate process verifies the claim’s veracity using the selected evidence and generates interpretable reasoning.

3.1 Self-supervised Denoising Evidence Retrieval

To ensure comprehensive evidence coverage for verifying the veracity of a given claim c , we retrieve candidate evidence from diverse and complementary sources:

Wikipedia-based retrieval. Following FEVER [38] and BEVER [5], we extract noun phrases from the claim via constituency parsing and use them to query Wikipedia articles. Sentences from

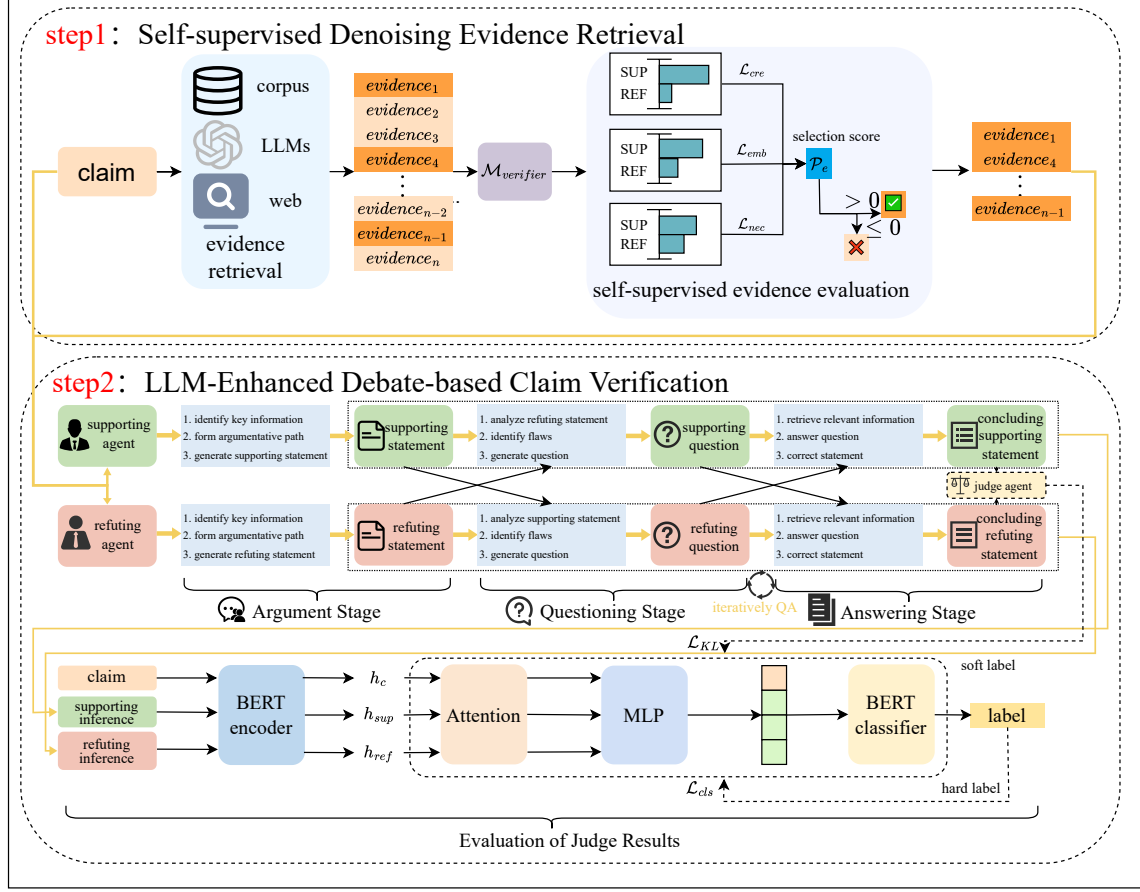


Figure 2: Overall framework of our proposed approach.

the top-5 articles are ranked using BM25 and BERT-based similarity, with the top-10 from each retained as candidates $\mathcal{E}_{wiki}$. **Google-based web retrieval.** Following CHEF [10] and CREDIFY [24], using the full claim as a query, we retrieve the top-5 Google search results. Candidate sentences are extracted and ranked using both BM25 and BERT-based models, denoted as $\mathcal{E}_{web}$. **LLM-based evidence generation.** Following GENREAD [51], we prompt large language models with standard and CoT-style instructions to generate two complementary evidence segments, denoted as $\mathcal{E}_{llm}$. The prompt can be referred in Appendix A.

Retrieved evidence from multiple sources often contains noise, which can hinder accurate verification. To address this problem, we first train an evidence selector $M_{verifier}$ based on the claims and their corresponding evidence. The trained selector $M_{verifier}$ is then employed to evaluate each retrieved piece of evidence in terms of its *credibility* and *necessity*, enabling the model to filter out irrelevant or misleading content while retaining essential evidence. Formally, given a claim c and a retrieved evidence set $\mathcal{E} = \mathcal{E}_{wiki} \cup \mathcal{E}_{web} \cup \mathcal{E}_{llm} = (e_1, \dots, e_n)$, the selection module identifies a refined subset $(e_1, \dots, e_k)$ ($k \leq n$) for downstream verification.

Specifically, given a claim c and a set of associated evidence sentences from the training data $(e_1, \dots, e_n)$, we concatenate them into a single input sequence, prepended by a special classification token [CLS]: $S_{emb} = \{[CLS] \oplus c \oplus e_{1:n}\}$. We use the hidden representation at the [CLS] position as the aggregated representation of the entire input sequence, which is then fed into the initial verification model $M_{verifier}$, which consists of BERT Encoder and multi-layer perception, to obtain its output representation:

$$O_{ce} = M_{verifier}(S_{emb}), \quad (1)$$

where O_{ce} denotes the predicted probability distribution over the $\{SUP, REF\}$, and SUP, REF respectively indicate that the evidence supports or refutes the claim. We then compute the cross-entropy loss between the predicted probability distribution and the ground-truth label distribution:

$$\mathcal{L} = \mathcal{L}_{CE}(O_{ce}, y^*), \quad (2)$$

where y^* denotes the one-hot encoding of the ground-truth label $y \in \{REF, SUP\}$. Then the $M_{verifier}$ is trained using gradient descent and backpropagation to minimize the cross-entropy loss.

After training the M_{verifier} , we use it to assess each retrieved evidence sentence from two perspectives: *credibility* and *necessity*. The *credibility* metric aims to quantify the individual contribution of a single piece of evidence to the veracity prediction, reflecting its independent influence on the model's decision. We can obtain the *credibility* representation as following:

$$S_{\text{cre}}(e_i) = \{[\text{CLS}] \oplus c \oplus e_i \oplus [\text{MASK}]^{(n-1)}\}, \quad (3)$$

where only e_i is visible while the remaining $n-1$ evidence slots are masked. While *credibility* metric captures the positive contribution of each evidence item, it fails to account for the relative indispensability of that evidence in comparison with others. To complement this, we define *necessity*, which measures the essential role of a given piece of evidence in assisting the veracity prediction. We can implement as following:

$$S_{\text{nec}}(e_i) = \{[\text{CLS}] \oplus c \oplus \tilde{E}_i \oplus [\text{MASK}]\}, \quad (4)$$

where $\tilde{E}_i$ denotes e_i is replaced with a [MASK] token in the evidence sequence, with all other sentences retained.

We then compute the cross-entropy loss for each evidence sentence based on its predicted *credibility* and *necessity* representations using the trained M_{verifier} :

$$\begin{aligned} \mathcal{L}_{\text{cre}} &= \mathcal{L}_{\text{CE}}(M_{\text{verifier}}(S_{\text{cre}}), y^*), \\ \mathcal{L}_{\text{nec}} &= \mathcal{L}_{\text{CE}}(M_{\text{verifier}}(S_{\text{nec}}), y^*). \end{aligned} \quad (5)$$

We compute the cross-entropy loss for whole evidence:

$$\mathcal{L}_{\text{emb}} = \mathcal{L}_{\text{CE}}(M_{\text{verifier}}(S_{\text{emb}}), y^*). \quad (6)$$

To avoid suboptimal evidence selection caused by noisy prediction errors of the claim verifier, we propose a significance-aware evidence evaluation strategy that assesses each evidence sentence based on its actual contribution to the claim verification. Specifically, we define a tunable threshold λ and retain evidence e_i only if the difference between its *credibility* and *necessity* loss, i.e., $\mathcal{L}_{\text{cre}} - \mathcal{L}_{\text{nec}}$, exceeds λ , indicating its significance in supporting or refuting the claim. However, when the verifier produces incorrect predictions, the embedding loss $\mathcal{L}_{\text{emb}}$ typically increases, suggesting unreliable evidence. To account for this, we introduce an adaptive thresholding mechanism that incorporates model confidence into the selection score:

$$\begin{aligned} \mathcal{P}_e &= \mathcal{L}_{\text{nec}} - \mathcal{L}_{\text{cre}} - \lambda \mathcal{L}_{\text{emb}}, \\ y' &= \begin{cases} 0, & \mathcal{P}_e \leq 0 \\ 1, & \mathcal{P}_e > 0 \end{cases}, \end{aligned} \quad (7)$$

where $\mathcal{P}_e$ is the evidence selection score and y' is the binary decision. This mechanism adaptively raises the selection threshold when the model exhibits low confidence, effectively filtering out misleading or uninformative evidence.

3.2 LLM-Enhanced Debate-based Claim Verification

After retrieving evidence, we propose a multi-agent debate framework to mitigate hallucinations in large language model reasoning caused by the omission of factual details in the claim. The debate process is divided into three stages: Argument Stage, Questioning

Stage and Answering Stage. The Questioning Stage and Answering stage are performed in an iterative manner, with the number of iterations denoted as R .

3.2.1 Argument Stage. Given the claim and the evidence retrieved during the retrieval stage, the supporting agent and the refuting agent independently construct arguments from supporting and refuting perspectives, respectively. Specifically, the construction process of each agent follows a Perception–Planning–Action paradigm: Perception focuses on identifying key informational elements within the claim (e.g., entities, dates, events); Planning involves organizing these elements in conjunction with evidence to form a structured argumentative path; and Action corresponds to generating a well-formed natural language statement that supports or refutes the claim in a logically consistent manner. The prompt template is included in Figure 3.

3.2.2 Questioning Stage. Upon receiving the response from the opposite agent, both the supporting and refuting agents increase the current debate round to $r + 1$. Each agent then identifies potential flaws, logical inconsistencies, or unsupported claims in the opponent's reasoning by jointly considering the opponent's argument and evidence, and formulates a targeted question intended to challenge or clarify the opponent's position. The prompt can be referred in Appendix B Figure 9.

3.2.3 Answering Stage. In this stage, both agents address their opponent's questions by retrieving relevant information from evidence and generating responses tailored to the specific inquiries. The Questioning Stage and Answering Stage proceed iteratively, and upon reaching the predefined number of rounds R , each agent generates a final concluding statement. The prompt can be referred in Appendix B Figure 10.

3.2.4 Judge Agent. Following the three-stage debate, we obtain analysis and reasoning grounded in the claim and evidence. To determine the final veracity prediction, a judge agent is prompted to evaluate the debate by assigning scores to the supporting and refuting agents, denoted as $(\text{Score}_{\text{sup}}, \text{Score}_{\text{ref}})$. The side with the higher score is selected as the final decision. The prompt can be referred in Appendix B Figure 11.

While LLMs are proficient at generating coherent reasoning, they often struggle to align this reasoning with claim labels in classification settings [22, 34]. In our debate framework, both supporting and refuting agents provide detailed arguments. However, relying solely on an LLM to interpret these arguments and determine the claim's veracity often proves inadequate [34]. To address this limitation, we introduce a training approach that integrates debate outcome evaluation with knowledge distillation. Specifically, the LLM judge's decision, along with the gold claim label, is distilled into a smaller model to improve its ability to synthesize and evaluate arguments from both perspectives.

3.2.5 Evaluation of Judge Results. The historical debate content of the supporting and refuting agents is concatenated as $T_{\text{sup}}, T_{\text{ref}}$ separately, and encoded alongside the claim c using a BERT encoder to obtain initial representations $h_{\text{sup}}, h_{\text{ref}}$, and h_c . To enhance the interaction between claim and debate content, inspired by Neural Semantic Matching Networks (NSMN) [25], we adopt a

The current scenario is a debate fact verification task, currently stage is [INITIAL STATEMENT].
 Given a claim and a set of retrieved evidence, your task is to analyze the evidence and construct a logical initial statement that supports the claim.
 The decision process follows a PERCEPTION–PLANNING–ACTION paradigm.

1. **Perception:** Identify key informational elements within the claim (e.g., entities, locations, dates).
2. **Planning:** Organize the identified elements in conjunction with the evidence to build a coherent argumentative structure.
3. **Action:** Generate a natural language statement that supports or refutes the claim logically and fluently.

(example)

CLAIM: Howard University Hospital and Providence Hospital are both located in Washington, D.C.
EVIDENCE: Providence Hospital is a 408 bed hospital located in the District of Columbia. . . . Howard University Hospital, previously known as Freedmen’s Hospital, is a major hospital located in Washington, D.C. . . .
Perception: The claim mentions two hospitals — Howard University Hospital and Providence Hospital — and asserts that both are located in Washington, D.C.
Planning: Examine the evidence for statements about the locations of these hospitals. Determine whether both are described as being in Washington, D.C. or its synonymous form, District of Columbia.
Action: Based on the evidence, generate a logically coherent supporting statement.
SUP_STATEMENT: The claim is supported by the evidence. Both hospitals are explicitly stated to be located in Washington, D.C. or the District of Columbia, which refer to the same geographic area. Therefore, the claim that Howard University Hospital and Providence Hospital are both located in Washington, D.C. is accurate.
 (more context examples . . .)

CLAIM:
EVIDENCE:
Perception:
Planning:
Action:

Figure 3: Example Prompt about Argument Stage of Supported Agent

multi-perspective interaction mechanism that jointly encodes both similarity and difference between claim representation h_c and debate content representations h_{sup}, h_{ref} :

$$\begin{aligned} e_{sup} &= h_c \circ (h_c \times h_{sup}) \circ (h_c - h_{sup}) \circ h_{sup}, \\ e_{ref} &= h_c \circ (h_c \times h_{ref}) \circ (h_c - h_{ref}) \circ h_{ref}, \end{aligned} \quad (8)$$

where $\circ$ denotes element-wise multiplication, and $h_c \times h_{sup}$ denotes the dot product, capturing semantic similarity between the claim and the supporting argument, while $h_c - h_{sup}$ encodes their semantic difference. Together, these operations form an interaction vector rich in semantic features. Additionally, an attention mechanism is employed to capture how the debate content from each side differentially responds to the claim:

$$\alpha_{sup} = \frac{\exp(\text{MLP}([h_c, h_{sup}]))}{\exp(\text{MLP}(h_c, h_{sup})) + \exp(\text{MLP}(h_c, h_{ref}))}, \quad (9)$$

$$\alpha_{ref} = \frac{\exp(\text{MLP}([h_c, h_{ref}]))}{\exp(\text{MLP}(h_c, h_{ref})) + \exp(\text{MLP}(h_c, h_{sup}))}. \quad (10)$$

The weighted representation of the debate content is then fed into a multilayer perceptron to obtain a relevance score:

$$\begin{aligned} score_{sup} &= \text{MLP}(\alpha_{sup} e_{sup}), \\ score_{ref} &= \text{MLP}(\alpha_{ref} e_{ref}). \end{aligned} \quad (11)$$

Finally, a softmax function is applied to the contrastive scores from both sides to produce a relative probability distribution over the debate outcome:

$$p_s = \text{Softmax}(score_{sup}, score_{ref}). \quad (12)$$

To enhance the small model’s ability to assess debate content, we employ knowledge distillation approach with both the ground-truth

label and the judge model’s scores as supervision. The ground-truth label y serves as a hard label, mapping (1, 0) for true/support and (0, 1) for false/refute, which guides the model via binary cross-entropy loss based on the agents’ reasoning:

$$\mathcal{L}_{cls} = \mathcal{L}_{BCE}(p_s, y), \quad (13)$$

where $\mathcal{L}_{BCE}$ denotes binary cross-entropy. This equation encourages the MLP and attention modules to prioritize reasoning from the side aligned with the ground-truth label. However, arguments from the opposing side can also enrich claim verification by offering alternative perspectives. To leverage this, we incorporate soft labels p^* derived from the judge agent’s score, providing additional supervision. A soft loss based on Kullback–Leibler (KL) divergence [3] between the model predictions and p^* is used to guide training:

$$\mathcal{L}_{KL} = D_{KL}(p^* || p_s) = \sum p^* \log \frac{p^*}{p_s}, \quad (14)$$

which encourages the MLP and attention modules to acquire partial knowledge from the LLM during evaluation. When the LLM’s predicted scores are uncertain, hard labels steer the model toward the correct classification. When the LLM is confident, soft labels help preserve its assessment. The final objective function $\mathcal{L}$ combines both losses to jointly optimize the encoder and prediction modules:

$$\mathcal{L} = \beta \mathcal{L}_{cls} + (1 - \beta) \mathcal{L}_{KL}, \quad (15)$$

where β is hyperparameter for balancing two losses. During inference, the final verdict of the claim is determined by selecting the side with the higher predicted probability based on the debate scores:

$$y = \operatorname{argmax}(p_s). \quad (16)$$

4 Experiments

4.1 Dataset and Evaluation Metrics

4.1.1 Dataset. Under the open-domain setting, we evaluate our approach on two benchmark datasets: the Chinese open-domain fact verification dataset CHEF [10] and the English multi-hop fact verification dataset HOVER [13]. The CHEF dataset consists of 9,001 training instances and 999 test instances, each labeled as one of three categories: SUPPORTS (SUP), REFUTES (REF), or NOT ENOUGH INFORMATION (NEI). For our open-book experiment settings, we focus only on instances labeled as SUP or REF. The HOVER dataset contains 18,171 training instances. Since its official test set is not publicly available, following [56], we use the development set as a substitute test set, which includes 4,000 instances comprising 1,126 two-hop, 1,835 three-hop, and 1,039 four-hop examples.

4.1.2 Evaluation Metrics. Following [10], we adopt micro-F1 and macro-F1 scores to evaluate verification performance on the CHEF dataset. For the HOVER dataset, we follow the evaluation protocol of [13], using macro-F1 to assess performance separately on two-hop, three-hop, and four-hop instances.

4.2 Baselines

We compare our method with two categories of baselines. The first includes fine-tuned pre-trained language models such as BERT and RoBERTa, which take the concatenation of the claim and evidence as input, where the evidence is obtained either by Pyserini toolkit (selecting the top-5 most relevant paragraphs from the indexed knowledge base) [21] or by our retrieval method, followed by a classification layer. The second includes LLM-enhanced approaches: Direct Prompting uses in-context examples for direct prediction; CoT Prompting guides LLMs to reason step-by-step; FOLK [43] converts claims into first-order logic clauses for guided reasoning; ProgramFC [27] decomposes claims into sub-tasks handled by specialized models; and MRR-FV [56] performs iterative evidence retrieval and multi-hop reasoning via memory tokens and a constructed reasoning graph. LoCal [21] proposes a logical and causal fact-checking framework based on multiple LLM-based agents.

4.3 Implementation Details

All experiments are conducted on a single NVIDIA RTX A6000 GPU with PyTorch 1.13.1. For evidence retrieval, we use bert-base-chinese and roberta-base-chinese on CHEF, bert-base-uncased and roberta-base-uncased on HOVER, with max input length 512, batch size 16, 10 epochs, and AdamW optimizer (learning rate $2e-4$, weight decay 0.01). For evidence generation, we use Deepseek-V3 [4] and Qwen-plus [36] on CHEF, and GPT-4o [26] and Qwen-plus on HOVER. In the verification stage, following LoCal [21], we adopt gpt-3.5-turbo as base agent. For in-context learning baselines, 20 prompt examples are used as in [28]. For debate evaluation and content training, we use bert-large with a max length of 1024, and AdamW (learning rate $5e-5$, weight decay 0.01).

Table 1: Comparison of different models on the CHEF. The best results are highlighted in bold, and the second-best results are underlined.

	Method	Micro-F1	Macro-F1
Pre-trained LMs	Pyserini + BERT	78.95	78.71
	Pyserini + RoBERTa	79.24	79.04
	Our evidence + BERT	82.58	82.35
	Our evidence + RoBERTa	84.13	83.86
LLM-enhanced	Direct	85.24	85.21
	CoT	86.66	86.72
	FOLK	87.22	87.17
	ProgramFC	88.75	88.45
	MRR-FV	89.23	89.07
	LoCal	<u>90.02</u>	<u>90.15</u>
	SLED (ours)	90.56	90.44

Table 2: Performance on the HOVER dataset by reasoning depth (2-hop, 3-hop, 4-hop). The best results are highlighted in bold, and the second-best results are underlined.

Category	Method	2-hop	3-hop	4-hop
Pre-trained LMs	Pyserini + BERT	63.79	53.26	49.51
	Pyserini + RoBERTa	64.66	54.65	51.61
	Our evidence + BERT	68.01	57.55	53.92
	Our evidence + RoBERTa	69.45	58.59	54.31
LLM-enhanced	Direct	60.51	54.75	53.61
	CoT	59.77	53.34	53.89
	FOLK	66.26	54.80	60.35
	ProgramFC	70.30	63.43	57.74
	MRR-FV	71.66	65.09	62.72
	LoCal	<u>72.71</u>	<u>64.11</u>	<u>61.59</u>
	SLED (ours)	73.69	66.92	64.66

4.4 Main Results

This section compares the performance of our proposed SLED model with several baseline methods on the CHEF and HOVER datasets. The results are presented in Table 1 and Table 2, respectively. Across both datasets, SLED consistently outperforms traditional retrieval and verification methods. Compared to evidence retrieved by Pyserini or the multi-hop retrieval method MRR-FV, evidence retrieved using our method leads to more accurate claim verification. This is attributed to our multi-source retrieval strategy, which enhances coverage and diversity, and our self-supervised denoising evidence retrieval module, which effectively filters out noisy information. These results demonstrate the effectiveness of our multi-source evidence retrieval and self-supervised denoising evidence retrieval framework.

In the comparison of claim verification strategies, we observe that supervised fine-tuned smaller models outperform LLM-based direct prompting and chain-of-thought (CoT) prompting on simpler two-hop claims. This may be due to potential conflicts between the LLM’s internal knowledge and retrieved external evidence, resulting in hallucinated reasoning [23]. However, for more complex three-hop and four-hop instances, LLM-based methods such as DebateFC, ProgramFC, FOLK, MRR-FV and LoCal demonstrate superior performance on complex claims compared to traditional fine-tuned models, emphasizing the importance of in-context learning and

multi-step reasoning in addressing challenging fact verification tasks. Compared with other LLM-based methods, our approach demonstrates superior performance on complex reasoning tasks by engaging in multi-round debates that iteratively identify key information. This indicates that guiding LLMs to focus on critical evidence can more effectively mitigate hallucination issues.

Table 3: Comparison of evidence retrieval strategies on the CHEF dataset. Ablation results demonstrate the contribution of $\mathcal{L}_{cre}$ and $\mathcal{L}_{nec}$.

Method	Micro-F1	Macro-F1
No evidence	63.25	63.41
Google Snippets	68.54	69.02
Surface Ranker	70.09	71.33
Semantic Ranker	71.31	70.15
Hybrid Ranker	74.79	75.21
Gold evidence	91.28	90.83
Our evidence	<u>82.58</u>	<u>82.35</u>
Our evidence (w/o $\mathcal{L}_{cre}$)	74.86	72.22
Our evidence (w/o $\mathcal{L}_{nec}$)	79.76	79.15

4.5 Evidence Retrieval Comparison

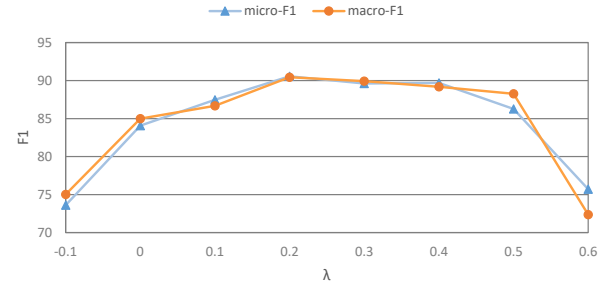
Since our approach incorporates search-based and LLM-generated evidence, comparing directly with gold evidence using F1 scores, common in previous work, is not sufficiently informative. Instead, we evaluate evidence quality by feeding evidence obtained through different retrieval methods into a BERT encoder with a multi-layer perceptron classifier trained on the CHEF training set. We then compare the resulting classification performance using micro-F1 and macro-F1 scores, as shown in Table 3. On CHEF, retrieval quality significantly affects verification performance. Methods using search engine snippets (e.g., Google Snippets) perform worse than those using curated documents (e.g., Surface Ranker, Semantic Ranker, Hybrid Ranker), indicating the importance of evidence source. Hybrid Ranker outperforms both Semantic and Surface Rankers, showing the benefit of combining retrieval strategies. Our method achieves results closest to using gold evidence, due to multi-source retrieval and self-supervised filtering. We also conduct ablation studies: **w/o $\mathcal{L}_{cre}$** (removing credibility loss) leads to over 7% performance drop, indicating the importance of reliable evidence selection. **w/o $\mathcal{L}_{nec}$** (removing necessity loss) causes a micro-F1 drop of 2.82% and macro-F1 drop of 3.2%, showing that necessity loss helps the model focus on relevant evidence.

Table 4: Ablation study on the impact of the debate evaluation and question-answering components in SLED across different reasoning depths on the HOVER dataset.

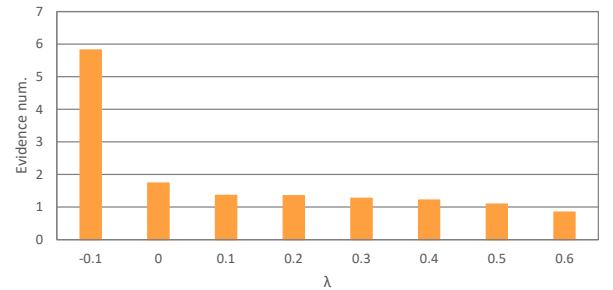
Method	2-hop	3-hop	4-hop
SLED (ours)	73.69	66.92	64.66
w/o debate evaluation	68.83	64.30	62.68
w/o question answering	66.86	59.13	57.21

4.6 Ablation Study on Claim Verification

We conduct ablation experiments to examine the impact of key components in our verification framework. **w/o Debate Evaluation** removes the debate evaluation module and directly uses the LLM-based judge’s decision as the final prediction. As shown in Table 4, performance drops across all claim types, with a 4.86% decrease on two-hop claims. This phenomenon suggests that large language models may be better suited for reasoning tasks but struggle to align with discrete labels in classification tasks, whereas fine-tuned smaller models tend to perform better on simpler classification problems. However, on three-hop and four-hop claims, performance remains relatively stable, indicating that LLMs themselves can better handle complex reasoning, supporting the effectiveness of using knowledge distillation to learn from LLM judgments. **w/o question answering** disables the questioning and answering between the two debating agents. Instead, the final judgment is based solely on their initial statements and evidence. This leads to performance drops of 5.17% and 5.47% on three-hop and four-hop claims, respectively. The decline reflects that, due to the agents’ comparable reasoning capabilities, their initial statements alone are often insufficient to differentiate a clear winner. The questioning stage and answering stage enable deeper interaction, exposing incomplete reasoning or weak evidence, which is critical for accurate claim verification in complex cases.

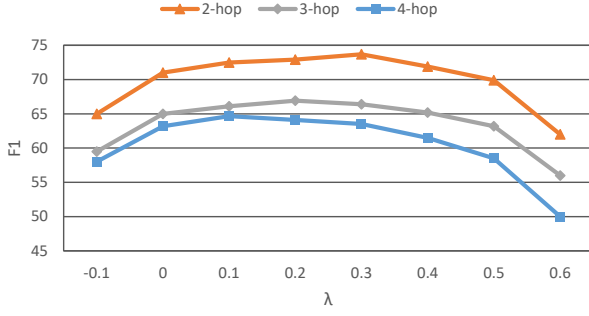
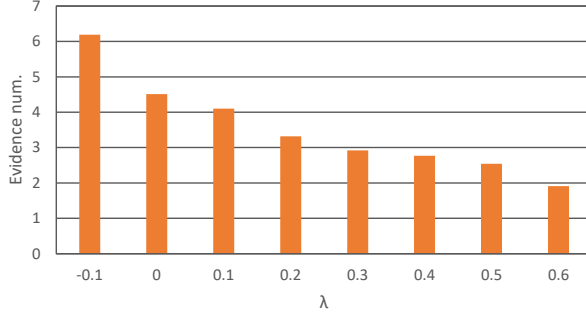


(a) Impact of the hyperparameter λ on micro-F1 and macro-F1 performance on CHEF dataset.



(b) Impact of the hyperparameter λ on evidence number on CHEF dataset.

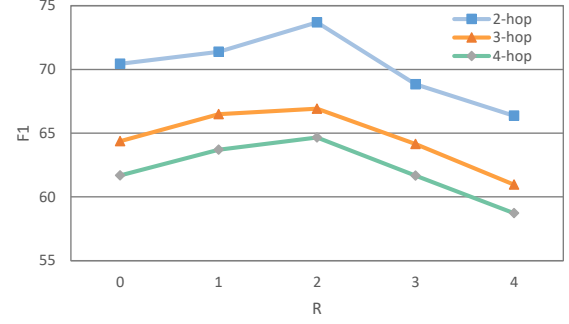
Figure 4: Impact of the hyperparameter λ on micro-F1 and macro-F1 performance and evidence number on CHEF dataset.

(a) Impact of the hyperparameter λ on F1 performance on HOVER dataset.(b) Impact of the hyperparameter λ on evidence number on HOVER dataset.**Figure 5: Impact of the hyperparameter λ on F1 performance and evidence number on HOVER dataset.**

4.7 Parameter Analysis

We conduct an analysis on hyperparameter λ , which regulates the evidence selection in Equation 7, to examine its effect on both micro-F1 and macro-F1 performance, as well as on the number of selected evidence in the CHEF and HOVER dataset. λ is varied from -0.1 to 0.6. The experimental results in Table 4 and 5 reveal a clear trend: as λ increases, the number of selected evidence instances decreases. When $\lambda = -0.1$, almost all evidence is selected, whereas at $\lambda = 0.6$, the average number of selected evidence items falls below one. Both extremes result in lower F1 scores, suggesting that the model’s performance degrades when it incorporates excessive noisy evidence or lacks sufficient supporting evidence. These findings underscore the importance of appropriate evidence selection in improving fact-checking accuracy. Additionally, the CHEF dataset primarily consists of one-hop and two-hop instances, whereas the HOVER dataset contains more complex cases involving two-hop, three-hop, and four-hop reasoning. The F1 scores on HOVER also vary with the number of selected evidence items, further demonstrating that selecting appropriate evidence is crucial for evidence-based fact-checking tasks.

To further investigate whether the improvements in multi-hop reasoning stem from the debate mechanism itself or from the repeated questioning stage and answering stage, we vary the maximum number of debate rounds R and analyze its impact. As shown in Figure 6, the best performance is achieved when $R = 2$, with consistent improvements across two-hop, three-hop, and four-hop

**Figure 6: Effect of the number of debate rounds R on F1 scores across different reasoning hop lengths.**

claims. When $R = 0$ (no questioning), performance drops, confirming the effectiveness of the debate interaction. However, when $R = 3$, performance declines again, especially on simpler two-hop claims. This suggests that additional debate rounds do not further enhance reasoning quality; instead, they may introduce ambiguity, making it difficult for the judge to discern the superior argument, thereby impairing final classification accuracy.

4.8 Error Analysis

We further conduct an error analysis by comparing our method with CoT on 25 correct and 25 incorrect cases for each model. Our method shows no hallucinations in correct predictions and only two in incorrect ones. Most remaining errors are caused by evidence retrieval failures (12 cases), NEI-leaning ambiguity (5), semantic misunderstanding (5), and reasoning errors (6). In contrast, CoT already exhibits hallucinations in correct cases (1) and a much higher number in incorrect cases (8), along with semantic misunderstanding (9), NEI-leaning ambiguity (3), and reasoning errors (5). Overall, these results indicate that the proposed debate mechanism effectively reduces hallucinations and promotes evidence-grounded reasoning. Representative cases are provided for illustration.

5 Conclusions

This paper presents SLED, a novel fact-checking framework that eliminate the noisy evidence in evidence retrieval and enhances veracity prediction through debate-enhanced reasoning. By incorporating self-supervised evidence selection and interactive verification with LLM agents, SLED effectively filters noisy evidence and reduces hallucinations in claim verification. Extensive experiments on CHEF and HOVER demonstrate that SLED achieves state-of-the-art performance, highlighting its potential for robust and scalable fact-checking.

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A Prompt of Generate Evidence

This section presents the prompts used to instruct ChatGPT to generate evidence, including the standard prompt in Figure 7 and the CoT prompt in Figure 8.

Standard Prompt Example for LLM-Based Evidence Generation
Instruction: Based on the given claim, use your retrieval capability to generate relevant evidence. Do not perform any judgment. Limit the response to no more than 180 characters. Ensure the evidence is directly related to the claim and provides the most persuasive information possible. Claim: <i>According to Chinese and foreign researchers, the COVID-19 virus had already been slowly spreading in both Asia and Europe before the fall of 2019.</i> Output: [Evidence]

Figure 7: Example of Standard Prompt for Generating Evidence.

CoT Prompt Example for LLM-Based Evidence Generation
Instruction: Fact-checking task: Given a claim, use your internal knowledge or external retrieval abilities to determine its veracity. From the information or evidence used in your judgment, extract the specific segments that directly support or refute the claim. Output these segments using the format ##Evidence 1, ##Evidence 2, etc. Claim: <i>According to Chinese and foreign researchers, the COVID-19 virus had already been slowly spreading in both Asia and Europe before the fall of 2019.</i> Output:

Figure 8: Example of CoT Prompt for Generating Evidence.

B Prompt Design for Debate Agents

This section presents the prompt designs employed during the multi-agent debate process. Specifically, Figure 9 provides the prompt used in the questioning stage, while Figure 10 shows the prompt template for the response stage. Finally, Figure 11 displays the prompt used by the judge agent to assess the debate outcome.

Prompt for the Questioning Stage of Supported Agent
The current scenario is a debate-based fact verification task. The current stage is [GENERATE QUESTION]. Assuming the claim is correct, as a supporting agent, use the provided evidence and your previous argument to critically question the refuting agent's statement. Claim: [CLAIM] Evidence: [EVIDENCE] Your statement: [SUP_STATEMENT] The refuting statement: [REF_STATEMENT] Actions: [QUESTION]

Figure 9: Example Prompt about Questioning Stage of Supported Agent

Prompt for the Answering Stage of Supported Agent
The current scenario is a debate-based fact verification task. The current stage is [QUESTION ANSWERING]. Assuming the claim is correct, as a supporting debater, based on the evidence, please rethinking and analysis your supporting statement, and answer the questions of the refute party, if current rounds equal to max rounds, generate a final statement supporting the claim. Claim: [CLAIM] Evidence: [EVIDENCE] Current rounds: [ROUNDS] Your statement: [SUP_STATEMENT] The refuting question: [REF_QUESTION] Notice: The generated content meets the factual standards. Actions: [ANSWER]

Figure 10: Example Prompt about Answering Stage of Supported Agent

Prompt for the Judge Agent
Q: As a fact checker, Please Scoring the performance based on records of the supporters and refuters's debate of the claim. Total of 10 points between two party, only generate the judgment score within two number of support:refute. [Supported Statement Records]/[Refuted Statement Records] A: [score]

Figure 11: Example Prompt about Judge Agent